

DFIR COMPLETE TOOLS REFERENCE GUIDE

Windows • Linux • Disk • Memory • Network • Timeline • SIEM • Threat Intelligence • Incident Response

A practical reference for learning what each major DFIR tool does, when to use it, where it fits in an investigation, and a recommended learning sequence.

1. The DFIR Tool Flow

ALERT → TRIAGE → ENDPOINT INVESTIGATION → EVIDENCE COLLECTION → DISK/MEMORY/NETWORK ANALYSIS → TIMELINE → IOC/THREAT INTELLIGENCE → CONTAINMENT → REPORTING

DFIR Stage	Primary Tool Types	Examples
Alert & Detection	SIEM/EDR	Wazuh, Splunk, Elastic, Sentinel
Endpoint Investigation	OS-native + endpoint tools	Event Viewer, PowerShell, Sysinternals, osquery
Collection	Artifact/evidence collectors	KAPE, Velociraptor, FTK Imager
Disk Forensics	Filesystem tools	Autopsy, Sleuth Kit, X-Ways
Memory Forensics	Memory analysis	Volatility 3, MemProcFS
Network Forensics	Packet/network analysis	Wireshark, Zeek, Arkime
Timeline	Timeline generation/analysis	Plaso, Timesketch
Threat Intelligence	IOC and enrichment	VirusTotal, MISP, OpenCTI
Detection Rules	Pattern/rule frameworks	Sigma, YARA
Case Response	Case management/SOAR	TheHive, Cortex, Shuffle

2. Evidence Collection & Endpoint Investigation

Tool	OS	Why / When Used
KAPE	Primarily Windows	Rapid collection of selected forensic artifacts
Velociraptor	Windows/Linux/macOS	Endpoint visibility, artifact collection and remote investigations
FTK Imager	Windows	Forensic image acquisition and examination
Magnet ACQUIRE	Windows	Evidence collection
osquery	Windows/Linux/macOS	Query endpoint state using SQL-like queries
GRR	Multi-platform	Remote live forensics and incident response
Sysinternals	Windows	Deep process, startup and system investigation

Typical flow: Alert → identify affected endpoint → collect relevant artifacts → preserve integrity → analyze copies → correlate with logs.

3. Windows DFIR Tools

Tool	Purpose	Use Case
Event Viewer	View Windows Event Logs	Login, account, service and security investigations
PowerShell	Query system and automate investigation	Collect system, process and event information
Process Explorer	Inspect running processes	Investigate suspicious processes
Process Monitor	Detailed system activity	File/registry/process behavior
Autoruns	Startup persistence inspection	Find persistence locations
TCPView	Network connection visibility	Identify process-to-network connections
Sysmon	Detailed telemetry generation	Process, network and file-related visibility
KAPE	Artifact collection	Rapid triage and collection

Tool	Purpose	Use Case
EvtxECmd	EVTX parsing	Windows log analysis
Timeline Explorer	Timeline review	Large CSV/structured timelines
Hayabusa	EVTX detection/hunting	Suspicious event detection
Chainsaw	EVTX hunting	Rule-based log hunting
Registry Explorer	Registry analysis	Analyze registry hives
RECmd	Registry extraction	Extract and process registry artifacts
PECmd	Prefetch analysis	Application execution-related artifacts
AmcacheParser	Amcache analysis	Application metadata/artifacts
JLECmd	Jump List analysis	User activity artifacts
LECmd	LNK analysis	Shortcut/file interaction artifacts

4. Linux DFIR Tools and Commands

Tool/Command	Purpose	Typical Investigation
journalctl	Query system journal	Service/system/security events
auditd	Security auditing	Track configured audit events
ausearch	Search audit events	Find relevant audit records
aureport	Audit reports	Summarize audit activity
ps	Process inspection	Suspicious running process
ss	Socket inspection	Listening/outbound connections
lsof	Open files/ports	Map process to resource
last	Login history	Interactive session history
lastlog	Last-login information	Account activity
osquery	Endpoint querying	System state and hunting
Velociraptor	Remote collection	Multi-host investigation

Useful sequence: authentication logs → login history → sudo/audit events → processes → sockets → persistence (cron/systemd) → file and configuration review.

5. Disk & File System Forensics

Tool	Purpose	When Used
Autopsy	GUI forensic analysis	Disk images and filesystem artifacts
The Sleuth Kit	Filesystem forensic utilities	Command-line analysis
FTK Imager	Acquire/examine images	Evidence acquisition
X-Ways Forensics	Advanced commercial forensics	Complex disk investigations
Magnet AXIOM	Commercial forensic suite	Artifact and device analysis
Arsenal Image Mounter	Mount forensic images	Controlled image examination

- Deleted-file recovery and examination
- MFT and filesystem metadata
- File timestamps and timeline context
- Browser and user artifacts
- USB/removable-media related artifacts
- File carving and partition analysis

6. Memory Forensics

Tool	Purpose	What You Analyze
Volatility 3	Memory forensic framework	Processes, modules, connections and memory artifacts
MemProcFS	Memory image access/analysis	Interactive memory examination
WinPmem	Memory acquisition	Capture RAM for later analysis
Dumplt	Windows memory capture	Acquire memory image

Memory principle: volatile evidence can disappear after shutdown/reboot. Follow organizational incident procedures and evidence-handling requirements.

7. Network Forensics

Tool	Purpose	Typical Evidence
Wireshark	Packet analysis	PCAP, protocols, sessions
tcpdump	Packet capture	Live/raw packets
Zeek	Network security monitoring	Structured network logs
Suricata	IDS/IPS detection	Alerts and network telemetry
Arkime	Large-scale packet capture/search	Session/packet investigation
NetworkMiner	Network forensic analysis	Hosts, sessions and extracted artifacts

Investigation questions: Who communicated? When? Which protocol? Which domain/IP? How much data moved? Was activity normal for the asset?

8. Timeline Analysis

Tool	Purpose
Plaso / log2timeline	Generate forensic timelines from many artifacts
Timesketch	Collaborative timeline analysis
Timeline Explorer	Visual analysis of structured timelines
SuperTimeline workflows	Combine normalized events from multiple sources

Example timeline: phishing → login → process execution → privilege change → persistence → internal connection → data access/exfiltration.

9. SIEM, Log Analysis and Detection

Tool/Platform	Purpose
Splunk	Search, correlation and SIEM investigations
Elastic Security	Search, analytics and security detection
Microsoft Sentinel	Cloud SIEM and security operations
Wazuh	Open-source security monitoring and detection
Graylog	Centralized log management
Hayabusa	Windows Event Log detection
Chainsaw	Fast EVTX hunting

10. Threat Intelligence & IOC Analysis

Tool	Purpose	Use
VirusTotal	Reputation and multi-engine analysis	Files, hashes, URLs and domains
MISP	Threat intelligence sharing/management	Indicators and threat context
OpenCTI	Threat intelligence knowledge management	Relationships and campaigns
YARA	Pattern-based matching	Identify malware/artifacts

Tool	Purpose	Use
Sigma	Generic detection rules	Translate detections across platforms
CyberChef	Data transformation/decoding	Decode, inspect and transform observables

Important: Reputation or intelligence results are supporting context, not automatic proof of compromise. Validate against your environment and evidence.

11. Incident Response & Case Management

Tool	Purpose
TheHive	Incident/case management
Cortex	Observable analysis and automation integrations
Shuffle	Security automation/SOAR workflows
ServiceNow Security Operations	Enterprise security incident workflow
Jira	Investigation tasks and coordination

12. Tools by DFIR Scenario

Scenario	Good Starting Tools	Analysis Path
Suspicious Windows login	Event Viewer, PowerShell, Hayabusa	EVTX → process → identity → timeline
Linux SSH compromise	journalctl, last, auditd	Auth → sudo → process → persistence
Malware alert	EDR + Sysinternals + KAPE	Alert → endpoint artifacts → timeline
Disk investigation	FTK Imager + Autopsy	Acquire → verify → analyze filesystem
Memory incident	Acquisition tool + Volatility 3	Capture → preserve → process/network analysis
Network exfiltration	Wireshark + Zeek + SIEM	Packets/flows → DNS → proxy → timeline
Enterprise hunt	Velociraptor + osquery + SIEM	Query fleet → collect artifacts → scope
IOC investigation	CyberChef + YARA + threat intel	Normalize → enrich → hunt/correlate

13. Recommended Learning Path

Phase	Learn These Tools First
1 — Foundation	Windows Event Viewer, PowerShell, journalctl, Wireshark, CyberChef, Sysmon
2 — Endpoint DFIR	Sysinternals, KAPE, Velociraptor, osquery, Hayabusa, Chainsaw
3 — Core Forensics	Autopsy, FTK Imager, Volatility 3, Plaso, Timesketch
4 — SOC + DFIR	Wazuh, Splunk/Elastic/Sentinel concepts, Sigma, YARA
5 — Advanced	MISP/OpenCTI, network monitoring, enterprise collection and threat hunting

14. Hands-On Tool Labs

Lab 1: Windows Event Investigation

Use sample or authorized lab EVTX data. Identify successful/failed authentication and create a short timeline.

Lab 2: Sysinternals Investigation

Use Process Explorer, Process Monitor and Autoruns on a lab VM to identify running processes and startup entries.

Lab 3: KAPE Collection

In an authorized Windows lab, collect selected artifacts and document what was collected.

Lab 4: Linux Authentication Investigation

Use journalctl and login-history commands to reconstruct an SSH event sequence.

Lab 5: Wireshark Analysis

Analyze a training PCAP: identify DNS, HTTP/TLS metadata and unusual communications.

Lab 6: Autopsy Disk Investigation

Open a training disk image and investigate files, metadata and selected artifacts.

Lab 7: Volatility Memory Lab

Analyze a known training memory image and identify process/network evidence.

Lab 8: Timeline Lab

Combine multiple evidence sources and build a chronological attack story.

Lab 9: Sigma/YARA Concepts

Create or examine safe detection patterns using known training samples.

Lab 10: Full Incident Challenge

Teams receive logs and artifacts, then produce evidence, timeline, scope and response recommendations.

15. Tool Selection Cheat Sheet

If your question is...	Start with...
Who logged in?	Windows Event Logs / Linux authentication logs
What process is suspicious?	Sysinternals / ps / EDR
How do I collect endpoint artifacts?	KAPE / Velociraptor
What happened on disk?	Autopsy / Sleuth Kit / FTK Imager
What was active in RAM?	Volatility 3
What happened on the network?	Wireshark / Zeek / SIEM
How do I build the story?	Plaso / Timesketch / Timeline Explorer
Is this IOC known?	Threat intelligence + CyberChef/YARA context
How do I manage the case?	TheHive / enterprise case workflow

16. Final DFIR Tools Architecture

DETECT → SIEM / EDR / IDS

INVESTIGATE → Event Logs / Sysinternals / journalctl / osquery

COLLECT → KAPE / Velociraptor / FTK Imager

ANALYZE → Autopsy / Volatility / Wireshark

RECONSTRUCT → Plaso / Timesketch

ENRICH → Threat intelligence / CyberChef / YARA / Sigma

RESPOND → Case management + containment + eradication + recovery

Golden rule: Do not learn 50 tools blindly. First understand the investigation question. Then choose the tool that collects or analyzes the evidence needed to answer that question.